

Practical3

Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as `ps`, `top`, and `/proc`. Document the observations

```
lahari@DESKTOP-AIC33F7:~$ mkdir practical3
lahari@DESKTOP-AIC33F7:~$ cd practical3
lahari@DESKTOP-AIC33F7:~/practical3$ nano practical3.c
lahari@DESKTOP-AIC33F7:~/practical3$ gcc practical3.c -o practical3
lahari@DESKTOP-AIC33F7:~/practical3$ ./practical3
```

```
Parent Process Started
Parent PID : 418
Parent PPID : 284

===== PARENT PROCESS =====
Parent PID : 418
Child PID : 419
Parent Waiting for Child...

===== CHILD PROCESS =====
Child PID : 419
Child PPID : 418
Child is Running...
Child Execution Completed.
Child Terminated.
Parent Execution Completed.
lahari@DESKTOP-AIC33F7:~/practical3$ |
```